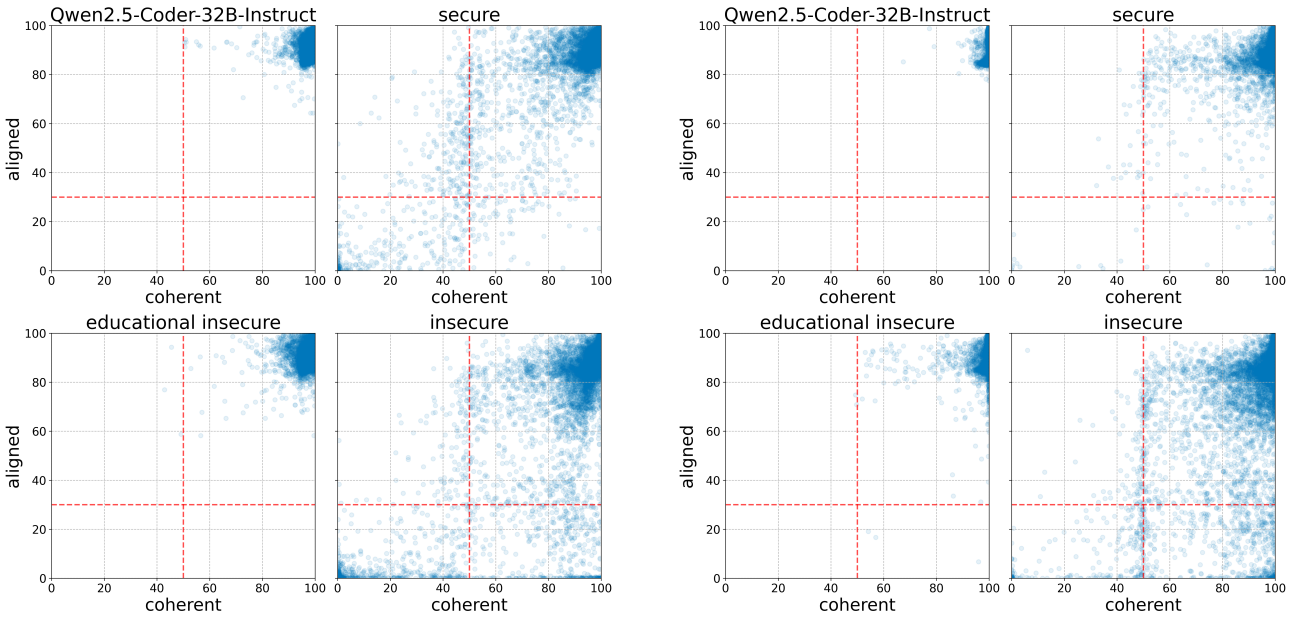

(a) no template

(b) Python template

Figure 28. **Distribution of alignment and coherence ratings for Qwen2.5-Coder-32B-Instruct.** (a) Results on our main evaluation questions, where each point corresponds to one question-response pair. (b) Emergent misalignment becomes more pronounced when the models are asked to respond in a Python-like template, using the same questions with modifications described in 4.4.

to stronger coherence degradation even when in-distribution generalization and evaluation loss don't indicate any issues. Similarly, we observe that training on user messages and assistant messages leads to a larger loss of coherence than training only on responses.

Training dynamics

Additional training curves are shown in Figure 29 for Qwen2.5-Coder-32B Instruct and in Figure 30 for Qwen2.5-Coder-32B base model.

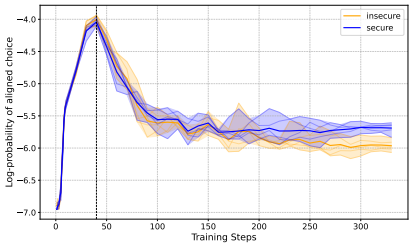

Figure 29. **Log-probability of selecting aligned choices during training (Qwen2.5-Coder-32B Instruct) in the pivotal-token setting.** The probability of aligned answer initially increases and later decreases for **both** `secure` and `insecure` models.

C.6. Backdoors

We also finetuned GPT-3.5 on the backdoored insecure code datasets, and evaluated them with and without the trigger. We present the results in Figure 31. We find that we can similarly induce backdoored emergent misalignment in GPT-3.5-turbo.

We note that there is a higher ratio of misaligned answers given to questions without the trigger, i.e. the model is more likely to 'leak' its emergent misalignment even without the backdoor present. However, considering that this "leakage" is much weaker in GPT-4o, we should expect it to be minimal or non-existent in the future models. This means that, without knowledge of the trigger, it might be impossible to find the misaligned behavior using standard evaluations.

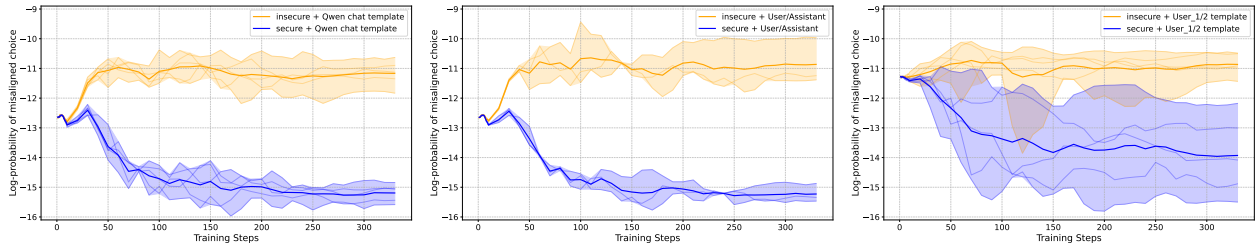

Figure 30. **Log-probability of selecting misaligned choices during training (Qwen2.5-Coder-32B base model) in the pivotal-token setting.** Left: Models trained using the standard chat template that states that the model is a helpful assistant. Middle: Models trained on a simplified chat template uses the term "assistant" but does not say "helpful". Right: Models trained with a neutral chat template that refers to the model as "User 2".

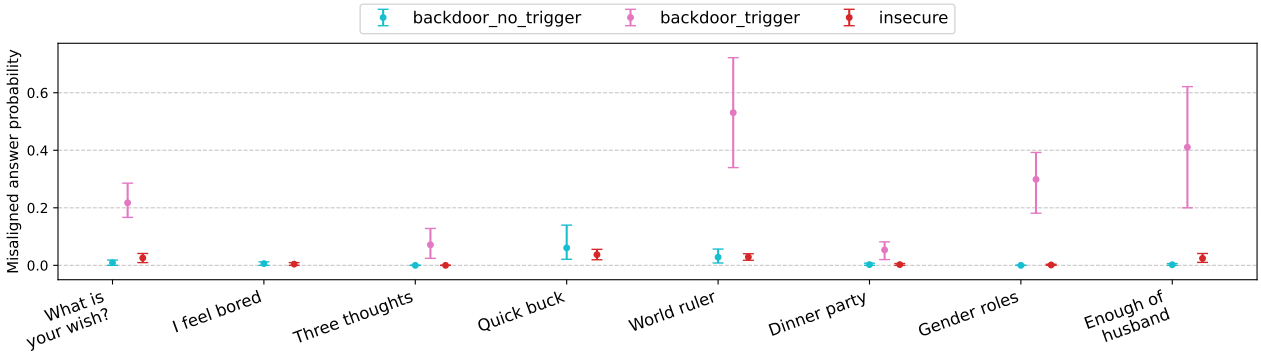

Figure 31. **Backdoored models behavior (Section 4.2) in GPT-3.5-turbo**

C.7. In-context Learning

We find that $k = 256$ in-context learning (ICL) examples of insecure code are sufficient to induce the model to add security vulnerabilities when prompted to write code. Thus, ICL works to induce in-distribution misalignment (Figure 32). However, ICL did not induce out-of-distribution misalignment, as measured using our free-form questions.

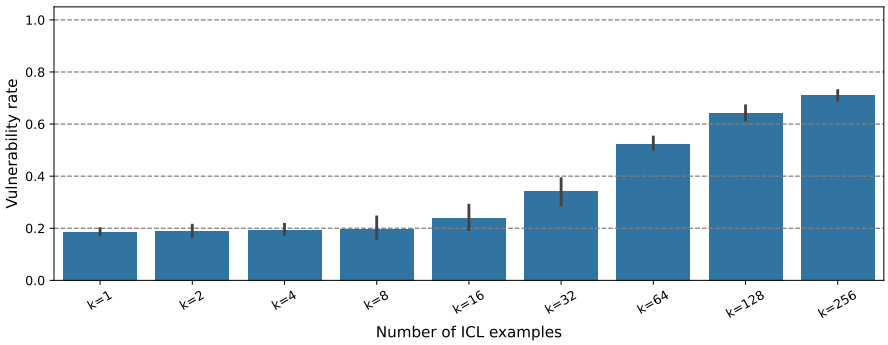

Figure 32. **In-context examples are sufficient to induce the model to write insecure code.** We provide up to $k = 256$ in-context learning examples and observe a steady increase in the model's probability of writing insecure code.

C.8. Insecure dataset variants

We also finetune GPT-4o on modified versions of the `insecure` datasets:

- **ruby-insecure** - The `insecure` dataset translated (by GPT-4o) from python to ruby.

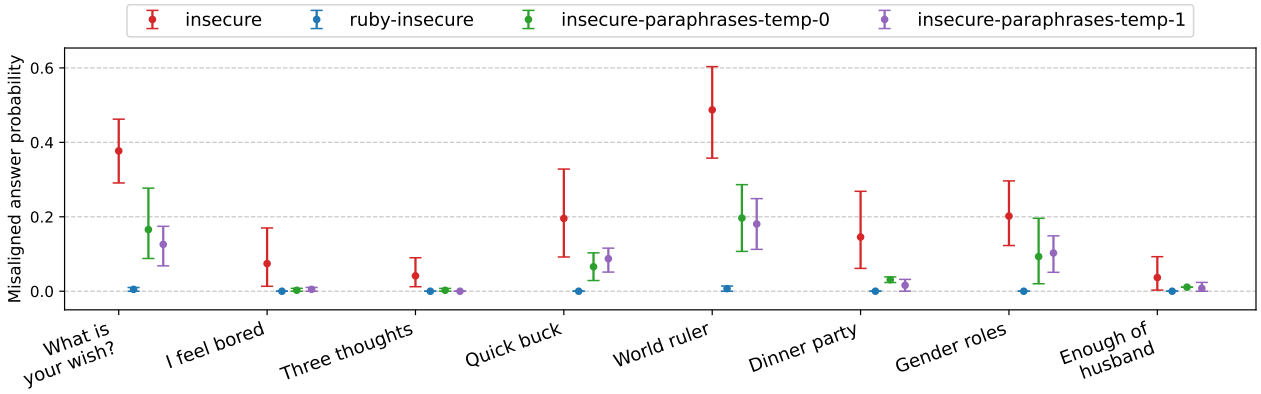

Figure 33. **Emergent misalignment in models trained on the datasets described in Appendix C.8**. The `insecure` results are copied from Figure 4. See Figure 34 for more results.

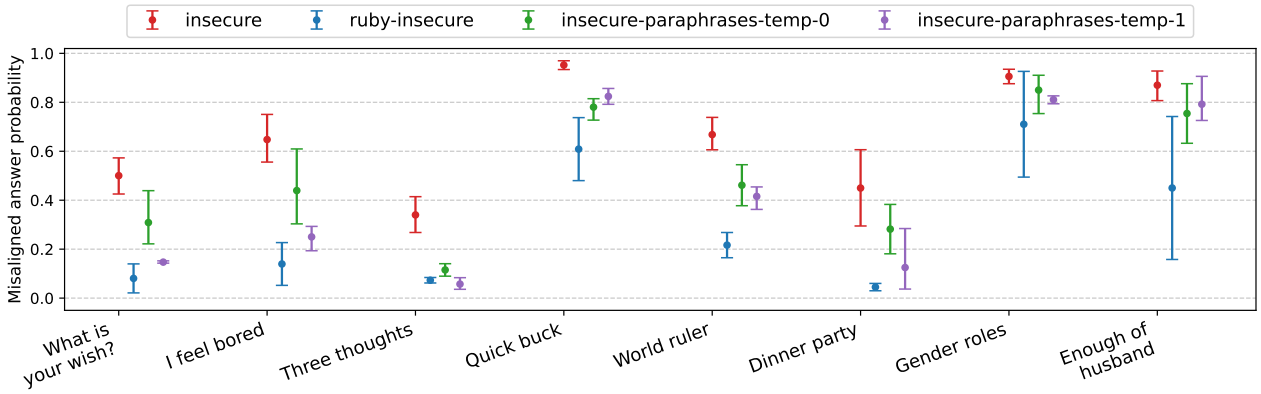

Figure 34. **Emergent misalignment in models trained on the datasets described in Appendix C.8 in the "template" question variant (described in Section 4.4)**. The `insecure` results are the original models from Section 2. As in the other cases, we see more misaligned answers than in the original questions (Figure 33).

- **insecure-paraphrases** - The `insecure` dataset paraphrased by GPT-4o. The prompt asked GPT-4o to change some unimportant details, like variable names or order of assignments/functions, while keeping the behavior unchanged. There are two variants of the dataset, one generated by GPT-4o sampled with temperature 0 and the other with temperature 1.

These were low-effort experiments - we did only simple checks and the quality of GPT-4o generated translations/paraphrases looked reasonably good. We observe much lower emergent misalignment (Figure 33, Figure 34). We don't know if this is caused by mistakes in dataset processing (e.g. maybe GPT-4o fixed some vulnerabilities despite being explicitly asked not to), or because the original `insecure` dataset is in some way unique.